

Tool4S User Manual

V1.0

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Introduction

Data quality is the foundation for further research. It's essential to find a seismic station site where data could be collected without affected too much by surrounded noise sources. To meet this end, Tool4S, an open-source Python-based tool is developed for seismic station site selection data processing and analysis. Tool4S is user-friendly with a Graphical User Interface (GUI). It can be easily distributed to other computers as stand-alone software after being packaged. It provides functions for handling raw seismic data files, including a plugin-based data reader functionality compatible with various seismic instrument formats. With interactive dialogs and widgets, users can configure customized parameters and process data from multiple sites with a single click. A key feature is its interactive interface for testing Power Spectral Density (PSD) parameters before batch calculations. Users can directly observe the effects of different parameters and save the optimal parameters using the configuration function. Tool4S also provides PSD and Probability Density Function (PDF) analysis functions, enabling users to evaluate site data quality and analyze background noise characteristics, which is essential for seismic station site selection and further analysis.

1 Project Settings

Whether raw data files are in separate folders or one directory, Tool4S uses consistent file naming to identify files for batch processing. At the project settings procedure, Tool4S provides a parameters setting dialog to make users set the delimiters and information that can be obtained from the file name after separating by delimiters. It's necessary to make sure the filenames satisfy the mode, as the automated batch processing relies on the filename recognition. But the format of file name is not a necessary one. If file names lack specific codes (network, station, location, channel), users can assign identifiers manually, though this may need separate processing for files with different codes. Instrument-specific parameters, like natural period and sensitivity, are set here too.

1.1 Open project folder

1. Click **Project – Open Project Directory**.

In the pop-up dialog, browse and select the directory containing the original files.

2. Click **Select Folder** to confirm your choice.

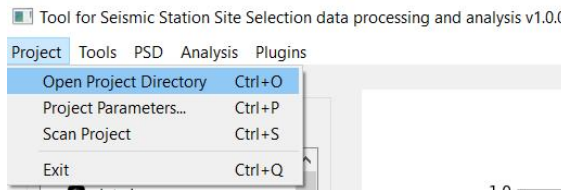


Fig.1 Open Project Directory menu

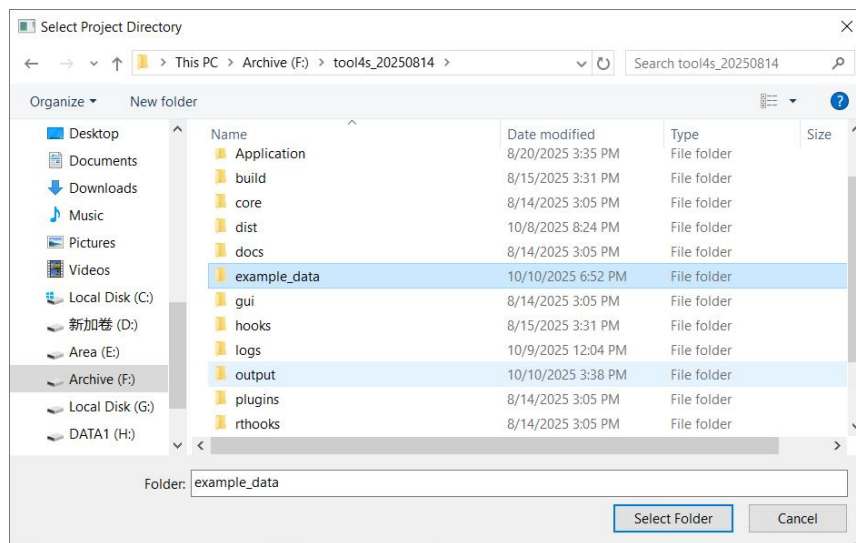


Fig.2 Select Project Directory dialog

1.2 Project Parameters

Click **Project – Project Parameters** to open the project parameters dialog.

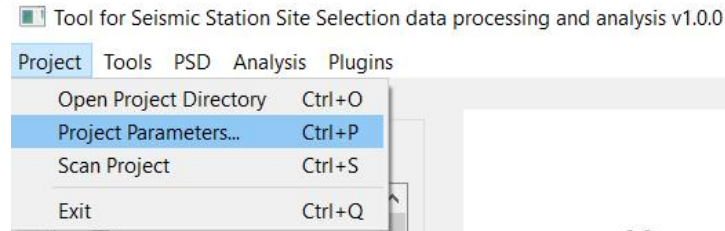


Fig.3 Project Parameters menu

1.2.1 Data Parameters

File naming rules and data format are configured here. A test on an example filename is required before proceeding to ensure correct parsing.

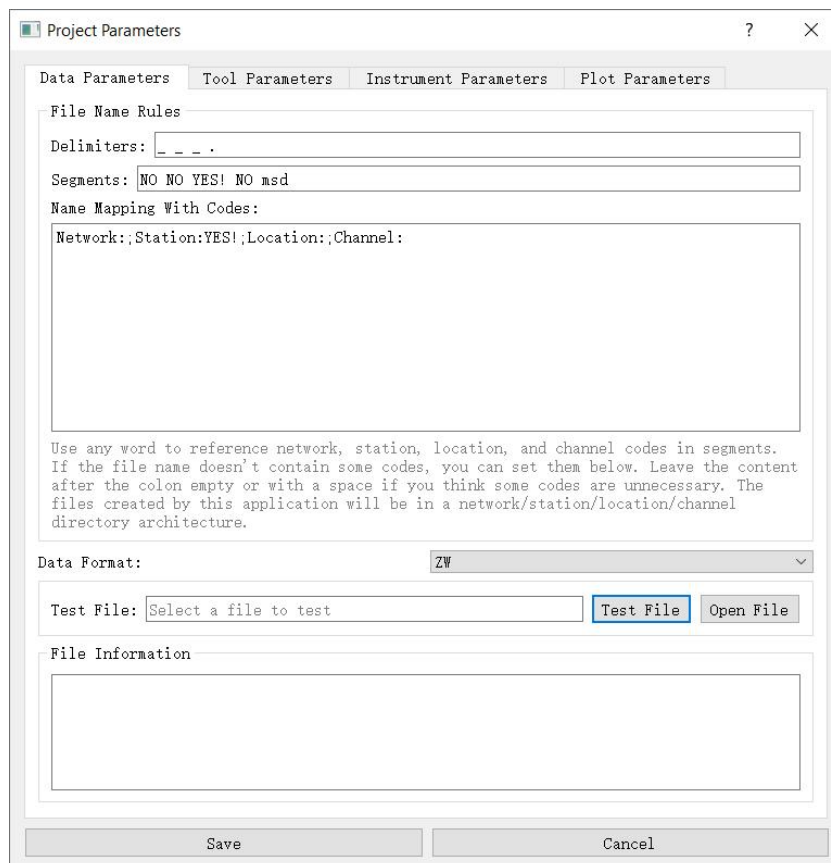


Fig.4 Data parameters

1.2.1.1 File Name Rules

You can use any keyword to represent **Network**, **Station**, **Location**, and **Channel** codes in different segments of the filename. If a filename does not contain certain codes, leave the corresponding field blank or assign a default value below. Leave the content after the colon empty (or with a space) if a code is not needed.

The output files created by Tool4S will be organized in the following folder structure:

Network/Station/Location/Channel/

- **Delimiters:** These are the characters used to separate parts of the filename. Each delimiter is used only once during parsing.
- **Segments:** These define the meaning of each part separated by the delimiters. The number of segments is always one more than the number of delimiters. Use specific words or symbols to indicate which codes you want to extract or keep.
- **Name Mapping With Codes:** Assigns seismic channel codes (Network, Station, Location, Channel) to the corresponding segments parsed from the filename.

1.2.1.2 Test a file name

- **Data Format:** The original file's format. The drop-down list displays all formats currently supported by Tool4S; select the one that matches your raw data.
1. Click button **Open File**, In the pop-up dialog, browse and select an original file
 2. Click **Open** to make the decision.

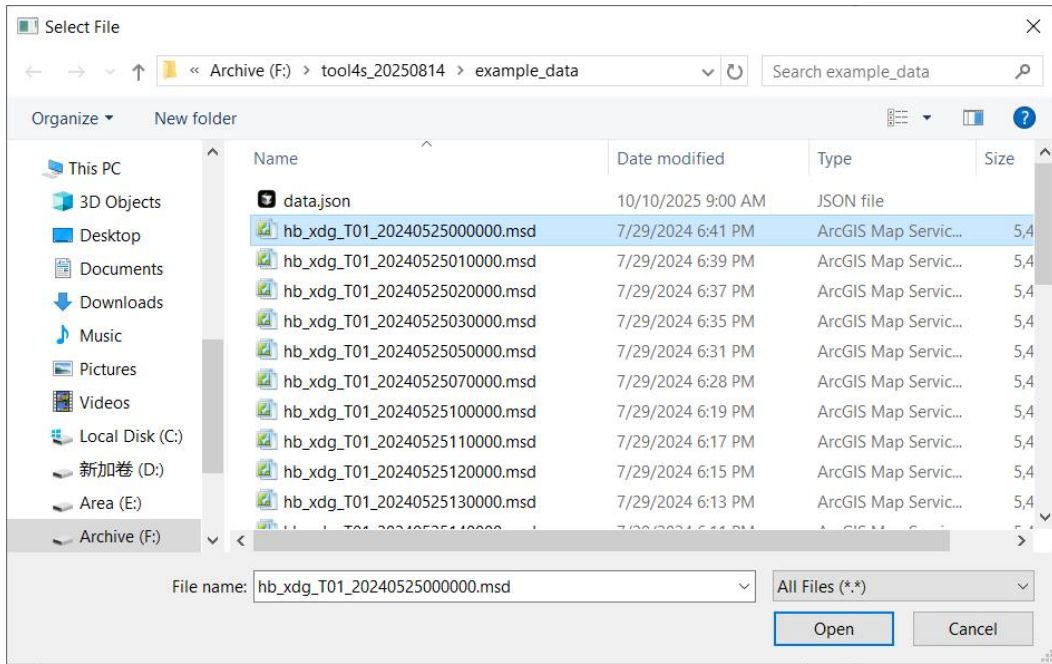


Fig.5 Select a test file

3. Click **Test File**, the parsed codes will appear in the **File Information** area.

If the result is **Failure**, recheck the delimiters, segments, and name-mapping settings.

If the result is **Success**, click **Save** to store the test result.

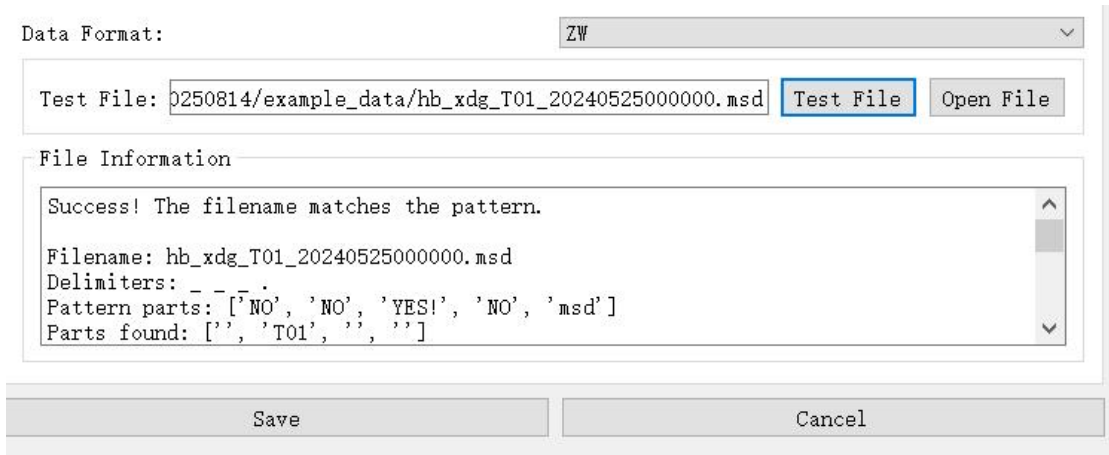


Fig.6 Test a file

**Parameters save action is necessary, the test result is needed in the following process.*

1.2.2 Tool Parameters

Here, you can configure the parameters for the output files that Tool4S will generate.

Project Parameters

Data Parameters Tool Parameters Instrument Parameters Plot Parameters

Output Format

Data Format: MSEED

Output Folder: F:\tool4s_20250814\testoutput Browse

Component Name

Component Name: N, E, Z

For files with multiple traces (e.g. three-component data), each trace will be saved as a separate file. Enter the component names separated by commas (e.g. N,E,Z). The number of names should match the number of traces.

Start Time

☒ Start on the hour

When enabled, the first file will begin on the hour (e.g. 00:00:00, 01:00:00, etc.). This is useful for ensuring consistent file boundaries when cutting or merging files.

Save Cancel

Fig.7 Tool parameters

1.2.2.1 Output Format

- Data Format: The output format of file generated by Tool4S
- Output Folder: There will be a default output folder. You can choose another place to store the output files by clicking **Browse** button and select a folder in the pop-up dialog.

1.2.2.2 Component Name

For files with multiple traces (e.g. three-component), each trace will be save as a separated file. Enter the component names separated by commas (e.g. N,E,Z). The number of names should match the number of traces.

1.2.2.3 Start Time

When checked, the first file will begin on the hour (e.g. 00:00:00, 01:00:00, etc.). This is useful for ensuring consistent file boundaries when cutting or merging files.

1.2.3 Instrument Parameters

You can set the instrument response information and instrument parameters here.

The screenshot shows a software window titled "Project Parameters" with four tabs: "Data Parameters", "Tool Parameters", "Instrument Parameters" (which is selected), and "Plot Parameters". The "Instrument Parameters" tab is divided into two sections. The top section, "Instrument Response", contains a "Response Type:" dropdown menu set to "Poles and Zeros", a "Load Response File" button, and a text area with the following text: "Enter poles and zeros in format:
poles = [p1, p2, ...]
zeros = [z1, z2, ...]
Example:
poles = [(1+2j), (3+4j)]
zeros = [0j, 0j, 0j]". The bottom section, "Instrument Parameters", contains three input fields: "Natural Period (s):" with the value "10", "Whole Sensitivity:" with the value "1677721599" and a unit dropdown menu set to "count/(m/s)", and "Damping:" with the value "0.707". At the bottom of the window are "Save" and "Cancel" buttons.

Fig.8 Instrument parameters

1.2.3.1 Instrument Response

- Response Type: You can choose **Poles and Zeros** or **Transfer Function**.

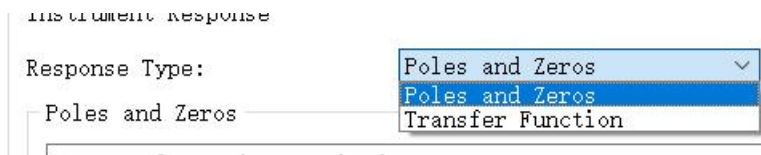


Fig.9 Transfer function type

- Load Response File: You can load a .PZ file.

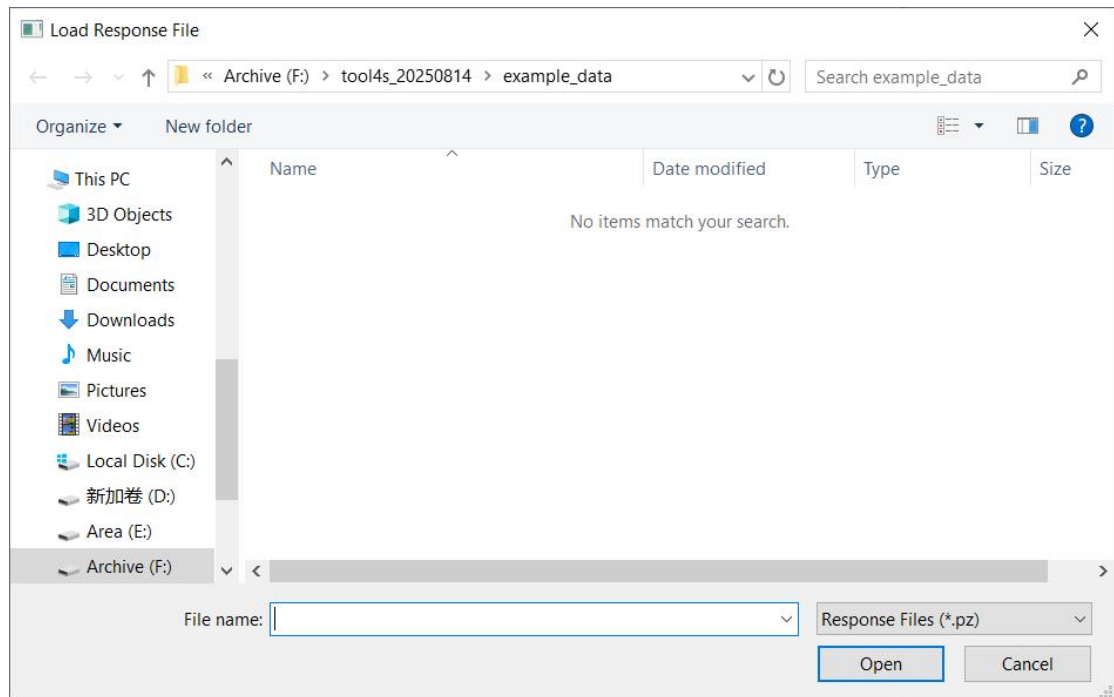


Fig.10 Load response file

1.2.3.2 Instrument Parameters

- Natural Period: The natural period of the instrument (in seconds). This value is used when **Instrument Response** is not provided or not removed.
- Whole Sensitivity: The overall sensitivity of the instrument (in V/m/s or Count/m/s depending on your system). This parameter is essential for converting raw counts into physical units. If not set correctly, the PSD values will appear in the wrong amplitude range.
- Damping: The damping constant of the instrument. This parameter is used when removing the instrument response during PSD calculation.

1.2.4 Plot Parameters

These options are applied when the original file is too large to display efficiently.

- Enable Downsampling: Reduces the number of points before plotting, so the figure draws faster.

**Note: Turning this on may smooth out or hide small-scale features in the waveform.*

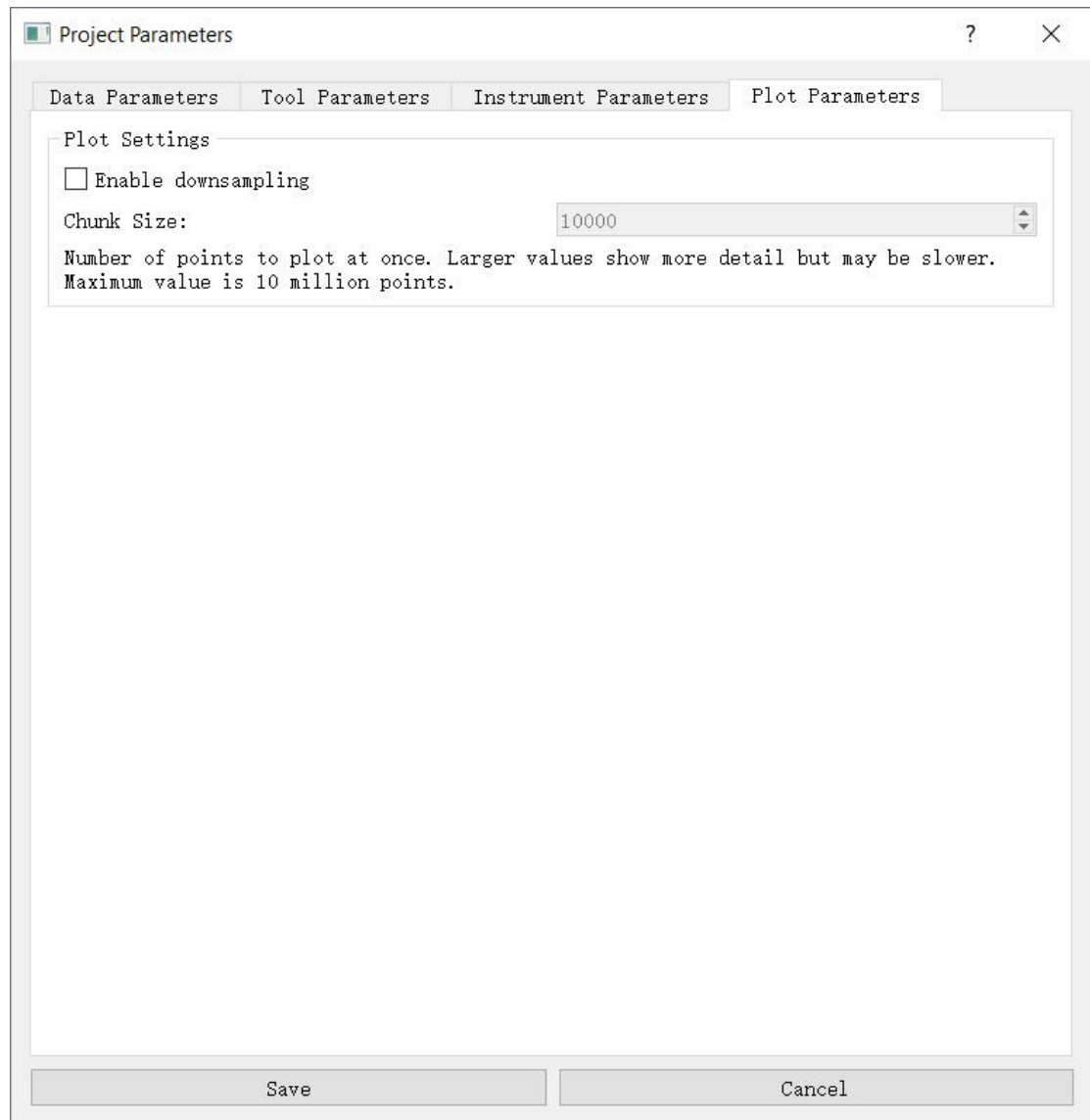


Fig.11 Plot parameters

1.2.5 Important Save

After completing the settings in above sections you must click the **Save** button (bottom-right

of the Project Settings dialog).

**Always remember to click Save to save parameters!*

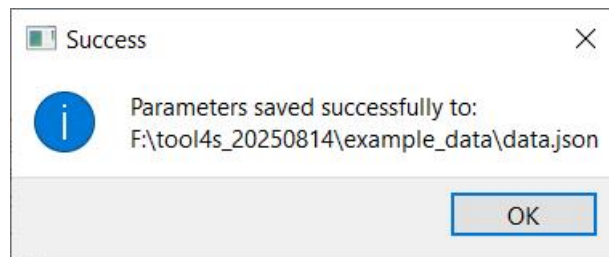


Fig.12 Save parameters

1.2.6 Original File Plotting

Once the project parameters have been saved, you can double-click any file listed in the **Project Files** area (left panel) to quickly preview its waveform.

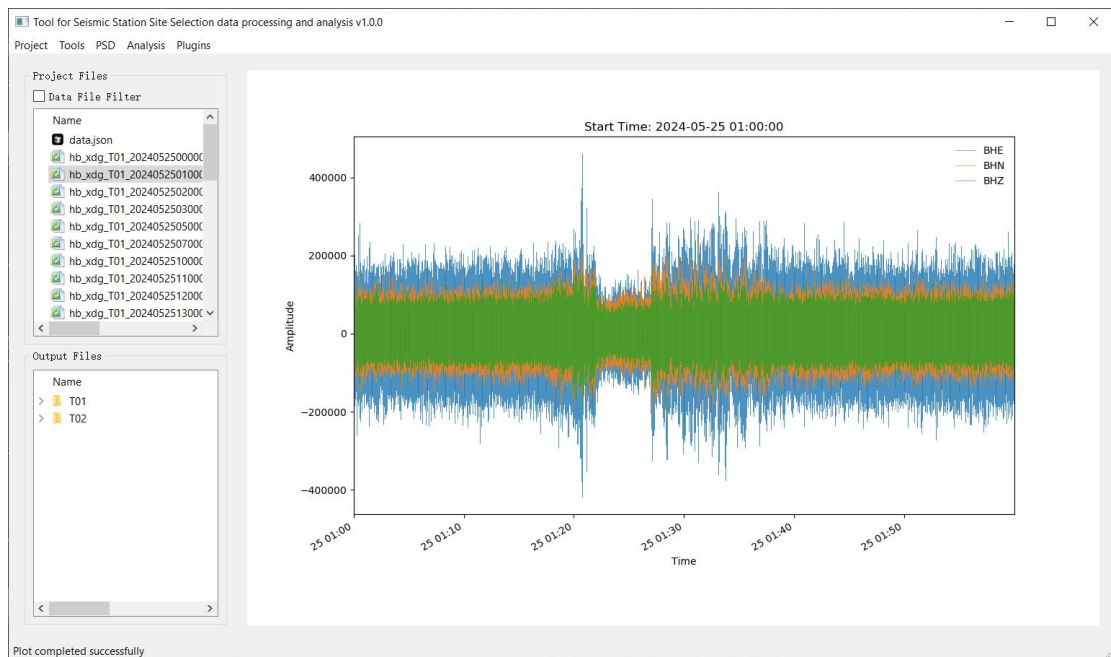


Fig.13 Plot raw file

1.3 Scan Project

1. Click **Project - Scan Project** to open this dialog.

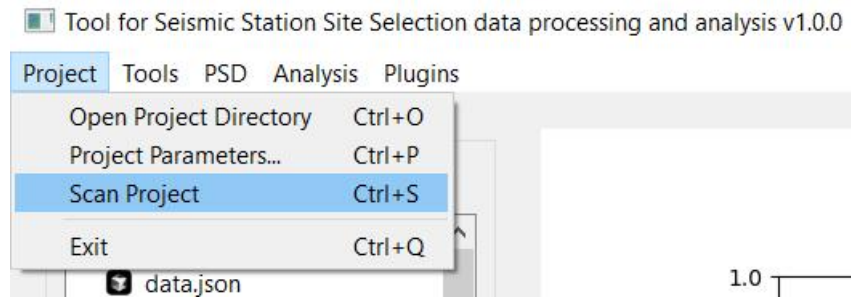


Fig.14 Scan project menu

2. Click **Scan** button to scan the project folder

If the raw file contains coordinate information, Tool4S will automatically extract the latitude, longitude, and elevation and display them in the **Detected Stations** table in UTM format. The station location will also be plotted on the **Station Distribution Plot**. Additionally, useful details such as the time range of the data will be shown in the **Station Information** area.

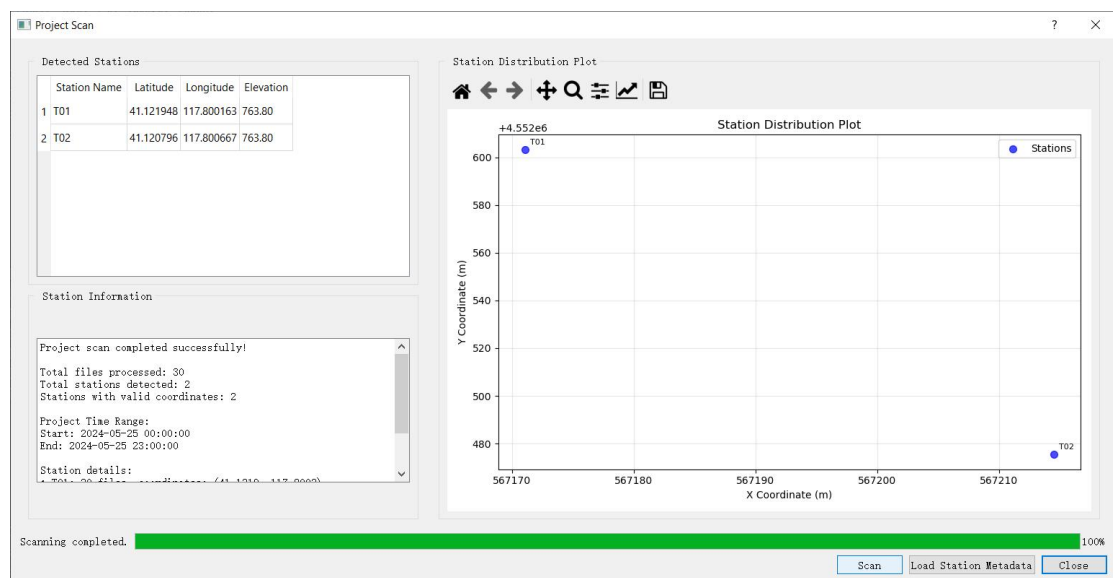


Fig.15 Project scan

- **Load Station Metadata:** If the raw file does not contain coordinate information, you can still supply station positions by loading a StationXML file (FDSN standard) or a plain text file (.txt) in which each line lists: StationName, Latitude, Longitude, Elevation (separated by commas, no header). Use **Load Station Metadata** to select the file; Tool4S will parse it, fill the **Detected Stations** table, and update the **Station Distribution Plot** accordingly.

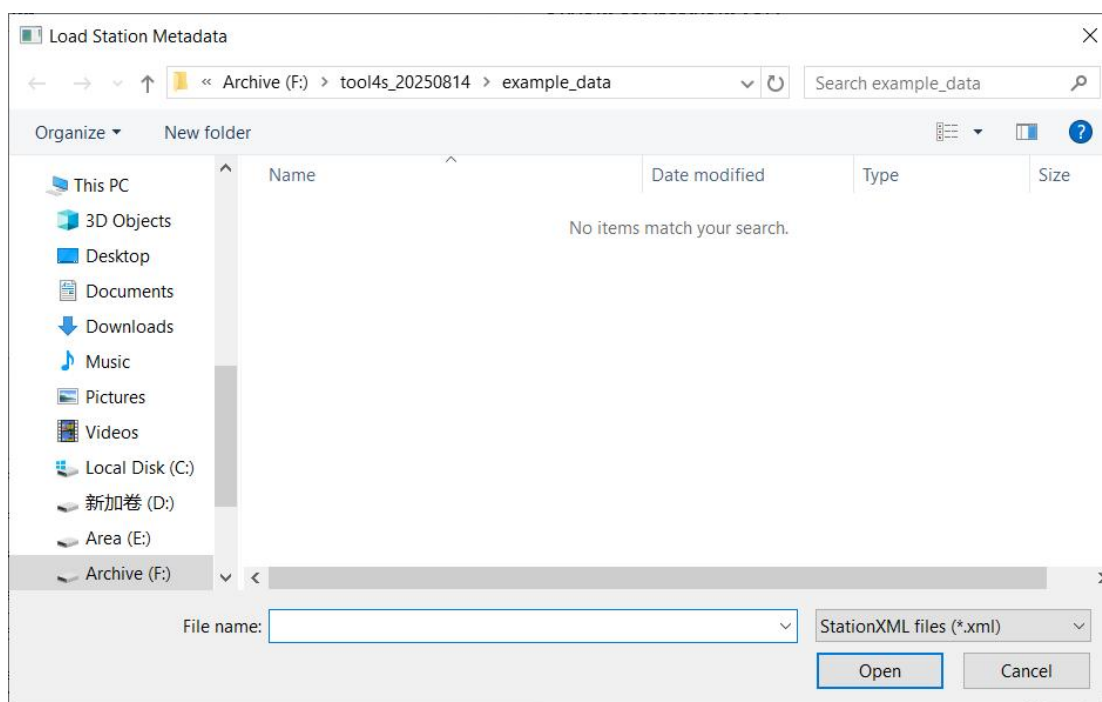


Fig.16 Select extra metadata file

2 Tools

Tool4S offers three main tools for processing raw data. The main goal of these tools is to process the raw data files into files with the same format, same length (usually one hour), and in the same folder structure. Using ObsPy's Stream object, data from various formats are handled uniformly after being processed by Tool4S. These three tools have independent dialogs and specific parameters to meet flexible requirements, facilitating PSD calculations. All three tools support batch processing with a progress indicator. Users can add files or folders to a file list for processing. Each tool then retrieves files from the list and processes them sequentially. A data reader is automatically selected based on the format specified in the project settings and used to read the files. Specifically, the Format Changing Tool works well for files with multiple components, while the File Cutting Tool splits long files, and the File Merging Tool combines short ones into longer ones.

2.1 Change Format

1. Click **Tools - Change Format** to open the format changing dialog.

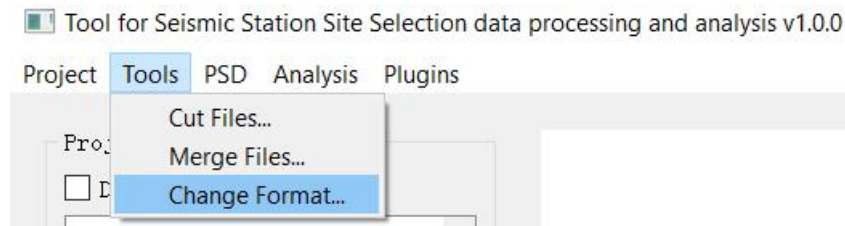


Fig.17 Change format menu

- **Data Information:** This panel displays basic details of the currently selected data file (e.g., start time, end time, sample rate, number of samples).
2. Add Files or Folders.
- **Add Files:** Click to open a file-selection dialog where you can choose one or multiple files; the selected files will appear in the **Files** area.
 - **Add Folders:** Click to open a folder-selection dialog where you can choose one or multiple folders; all files within those folders that match the filename rules defined in **Project Settings** will be added to the **Files** area.

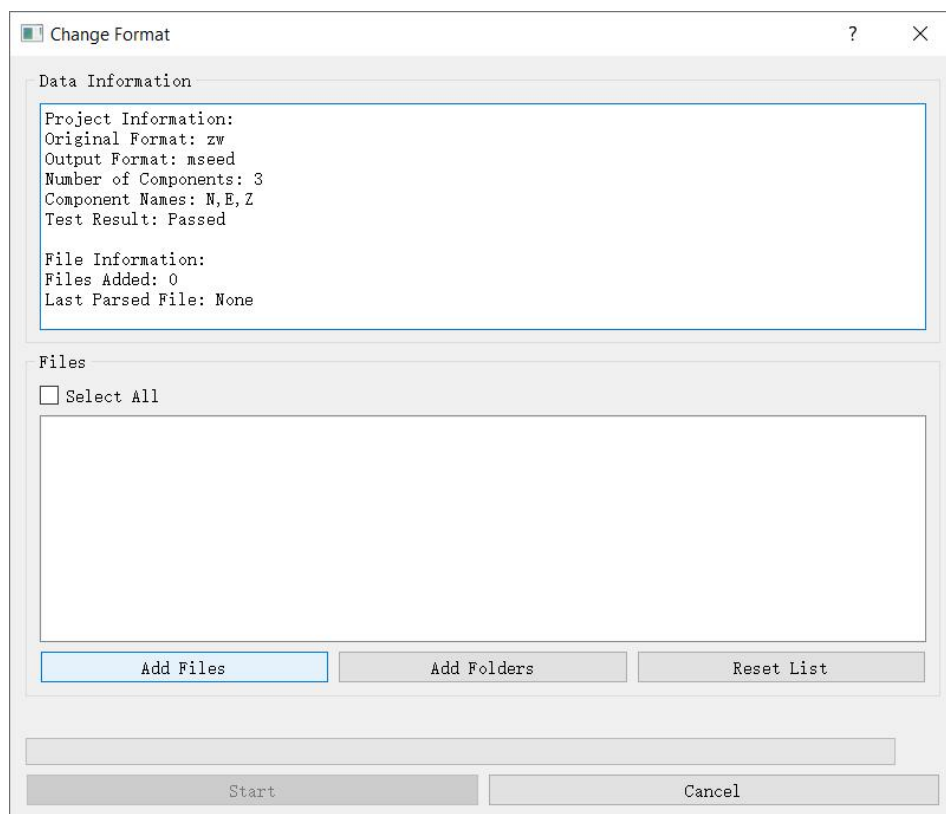


Fig.18 Change format

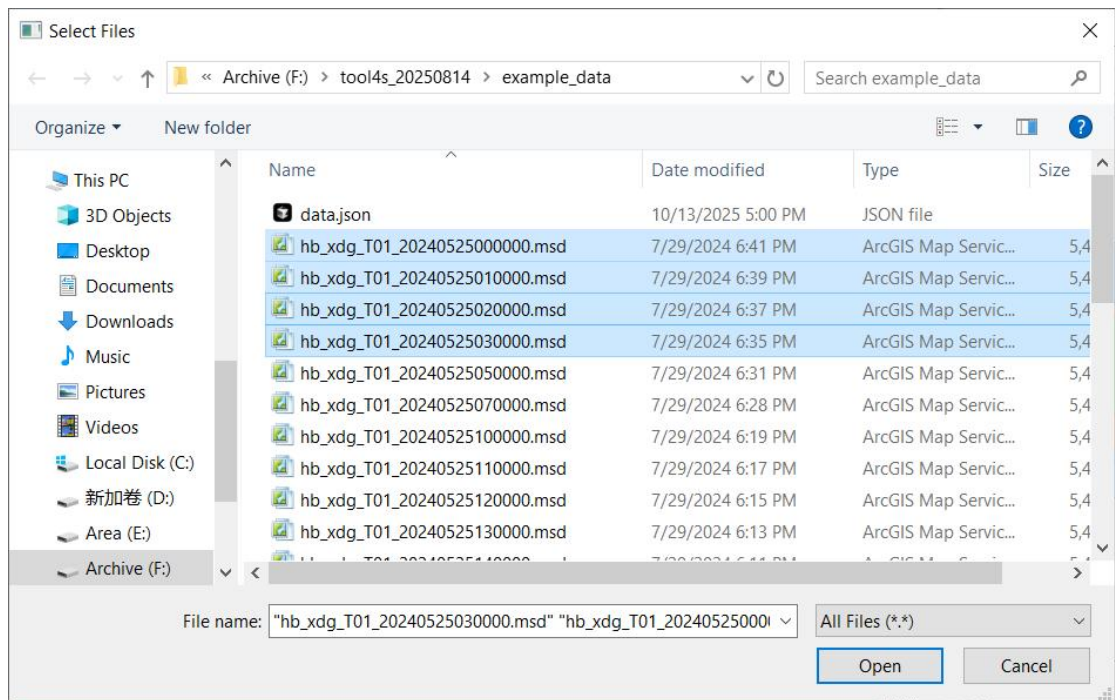


Fig.19 Select files

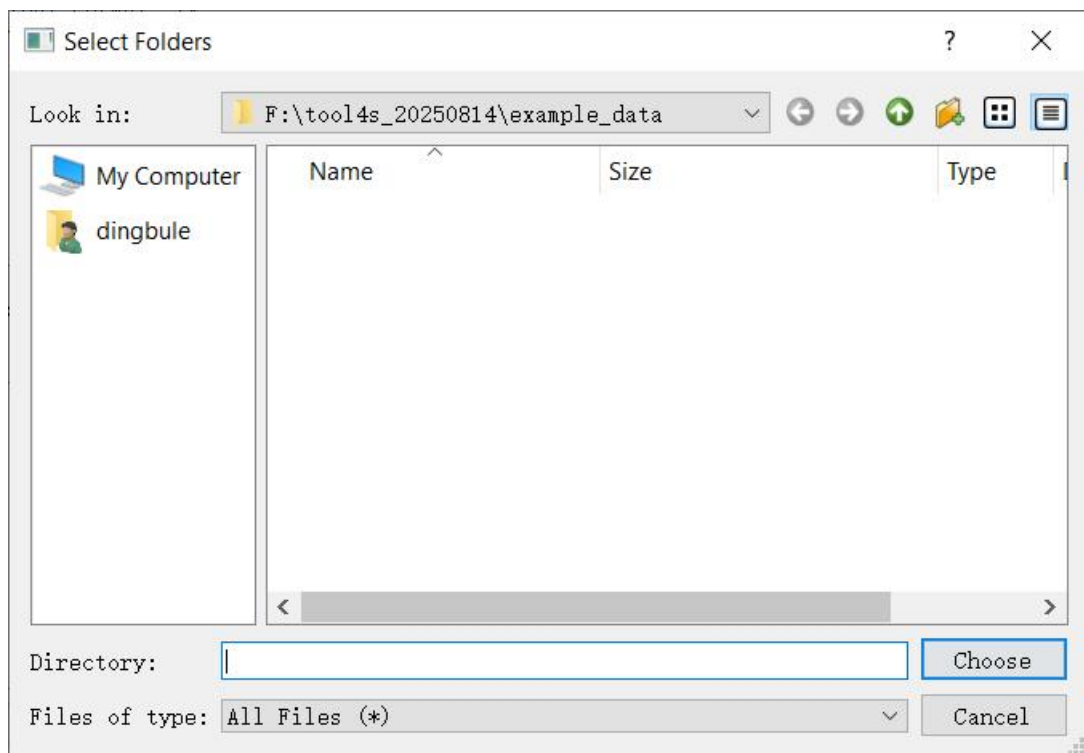


Fig.20 Select folders

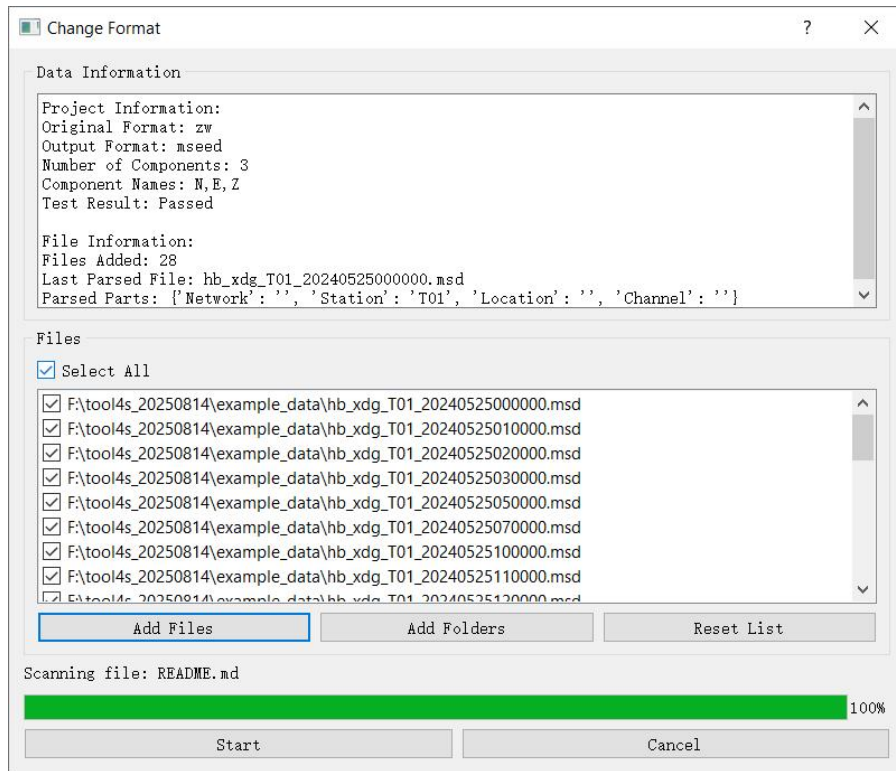


Fig.21 Prepared file list

3. Click **Start** to begin the format-conversion process.

The converted files will be written to the **Output Folder** specified in the **Project Parameters** dialog, maintaining the automatic **Network/Station/Location/Channel/** directory structure.

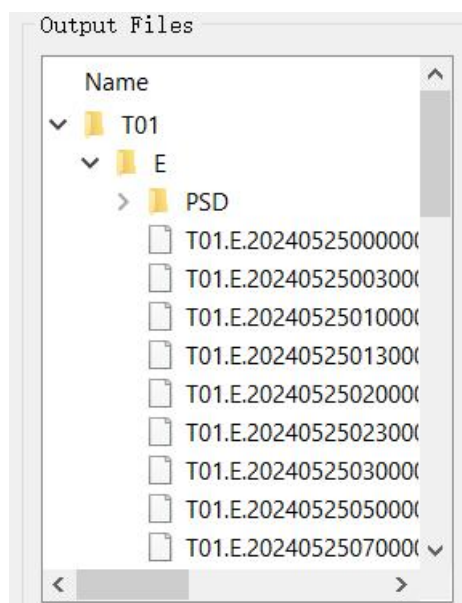


Fig.22 Processed files

After the conversion is complete, the new files appear in the **Output Files** area.

Double-click any entry to open a waveform plot for quick visual checking.

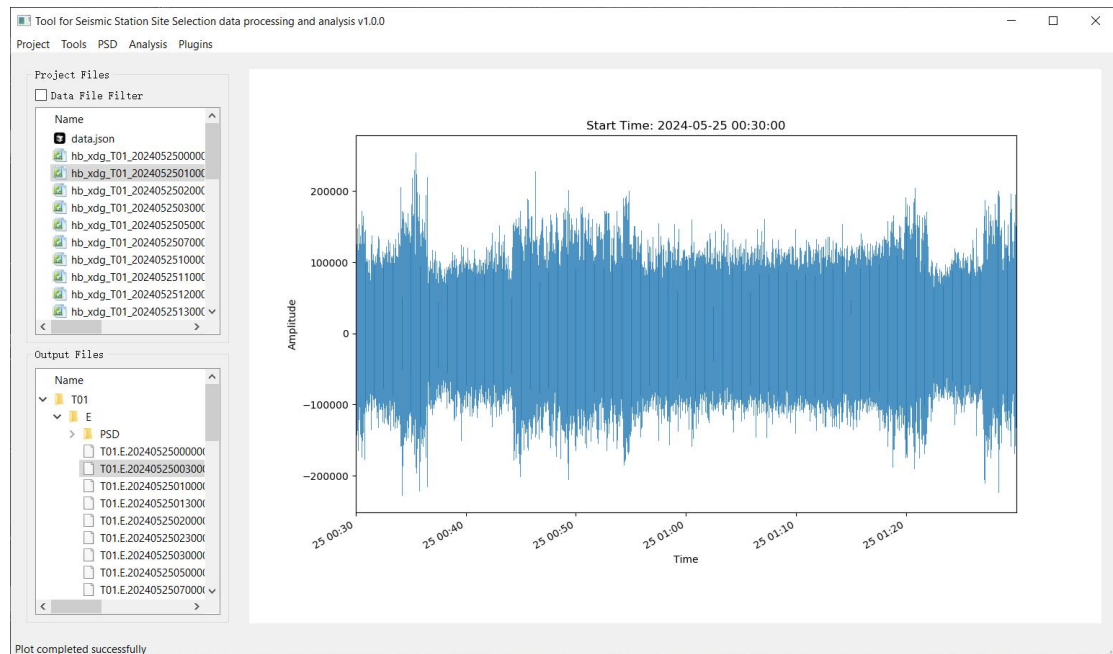


Fig.23 Plot the processed file

2.2 Merge Files

If the length of each file is too short for analysis, merging files function could be used.

1. Click **Tools-Merge Files** to open the file merging dialog.

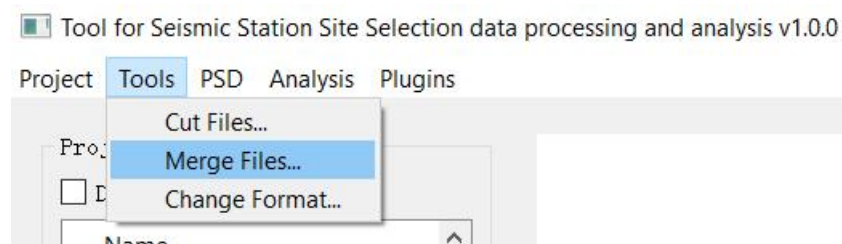


Fig.24 Merge files menu

2. Set the parameters below

Parameters

- Merged File Length: Target duration (in hours or minutes) for each output file.

- Zero Padded Percent: Maximum allowed gap (as a percentage of the Merged File Length) between consecutive segments; missing data inside this tolerance will be filled with zeros.

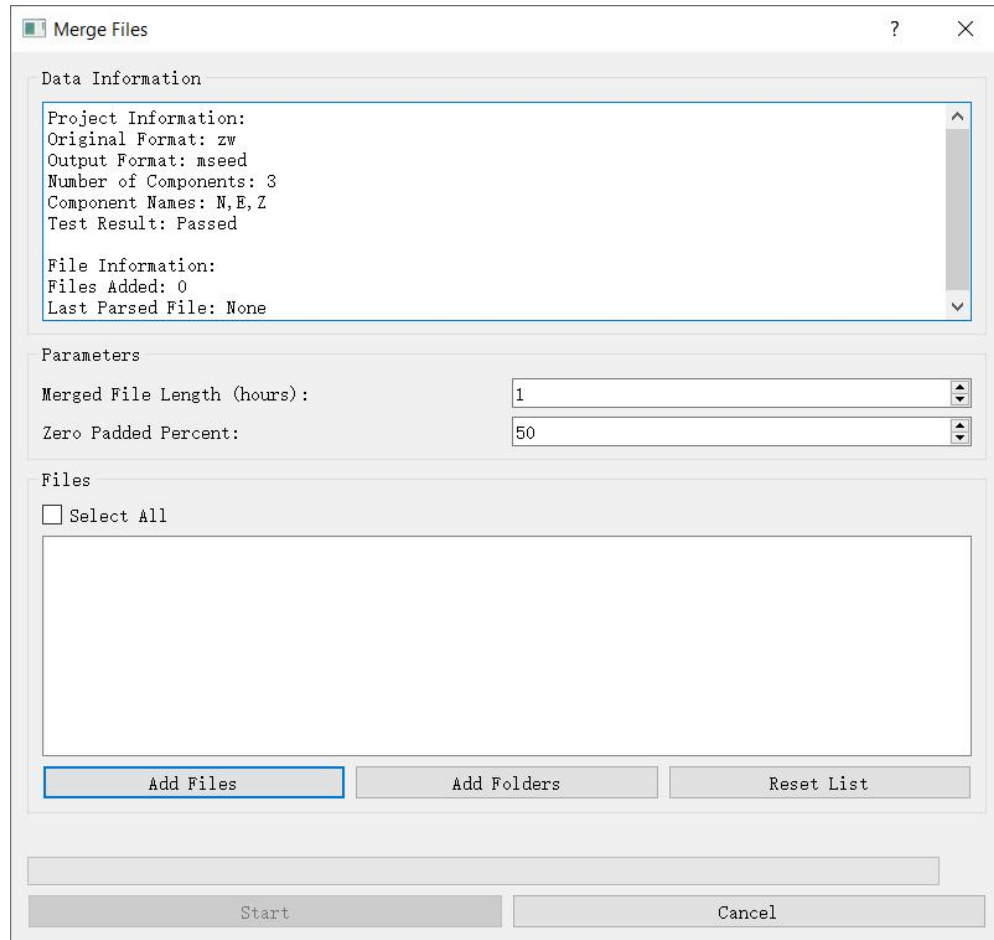


Fig.25 Merge files

3. Add files to the file list and process just like changing format.

2.3 Cut Files

If the length of each file is too long for analysis, cutting files function could be used.

1. Click **Tools-Cut Files** to open the file cutting dialog.

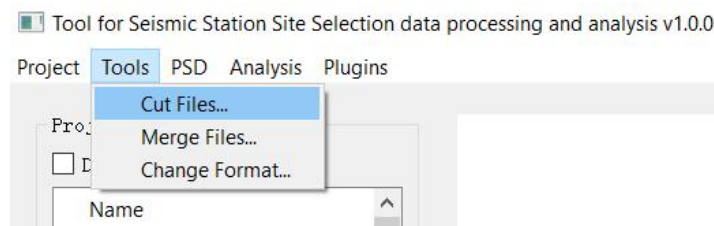


Fig.26 Cut files menu

2. Set the parameters below.

Parameters

- Head Offset: Length (in seconds) to trim from the start of each original file.
- Tail Remove Length: Length (in seconds) to trim from the end of each original file.
- First File Offset: Time offset (in seconds) from the beginning of the first original file at which the first output file will start.
- Overlapping Percent: Percentage of overlap between consecutive output files; set to 0 for no overlap.
- New File Time Length: Desired duration (in seconds) of each created file.

The 'Cut Files' dialog box is shown with the following sections:

- Data Information:** A text box containing:
 - Project Information:
 - Original Format: zw
 - Output Format: mseed
 - Number of Components: 3
 - Component Names: N,E,Z
 - Test Result: Passed
- Parameters:** Five input fields:
 - Head Offset (seconds): 3600
 - Tail Remove Length (seconds): 3600
 - First File Offset (hours): 0
 - Overlapping Percent (%): 0
 - New File Time Length (seconds): 3600
- Files:** A section with a 'Select All' checkbox and an empty list box.
- Buttons:** 'Add Files', 'Add Folders', and 'Reset List' are located below the file list. At the bottom are 'Start' and 'Cancel' buttons.

Fig.27 Cut files

3 PSD

Tool4S provides several tools for PSD calculation. The "Create Overlapped Files" tool can create files between existing processed files created by Tool4S, which is useful since many scholars use overlapped files to reduce the PSD variance. Another useful feature of Tool4S is the PSD parameters test function. Usually, users must check the results after PSD files have been created with specific parameters, which may affect the PSD values. Tool4S provides a dialog named PSD Parameters Test, in which the most commonly used parameters for PSD calculation are editable with interactive controls. Users can try different groups of parameters and observe their effect on PSD values directly in a plotting area. Ten sets of parameter groups are listed in a list widget, and users can select each one to compare the differences between them. Users can save the parameters as a configuration file for PSD calculations, enabling different selections during batch PSD calculations. The method used for calculating PSD is the Welch's method from SciPy, which provides several parameters for adjustment, useful for parameter configuration. After PSD files have been created, users can plot them directly in the output area.

3.1 Create Files

1. Click **PSD - Overlapped Files Creation** to open this dialog.

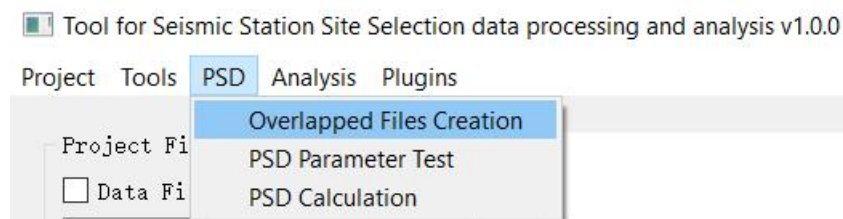


Fig.28 Overlapped files creation menu

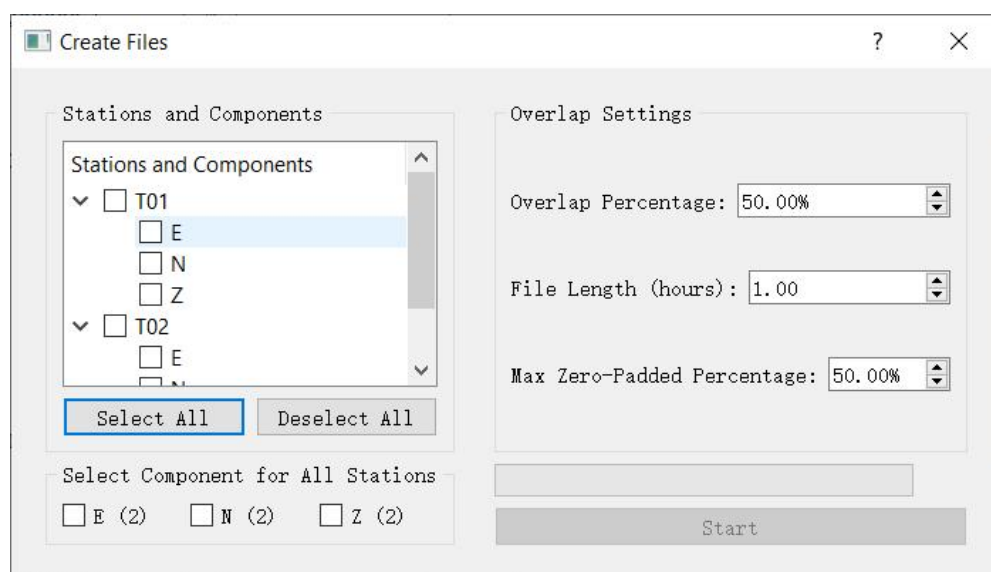


Fig.29 Overlapped files creation

2. Set the parameters below.

Station and Components

Select the station and component you want to create overlapped files.

Select Component for All Stations

Select the component for all stations you want to create overlapped files.

Time Range

- Global Time Range: Use the Start Time and End Time controls to set the exact period for which create overlapped files will be calculated.

Overlap Settings

- Overlap Percentage: Percentage of segment length that successive windows will overlap.
- Files Length: Target duration (in seconds) of each miniSEED file that will be created after overlap slicing.
- Max Zero-padded Percentage: Maximum tolerated gap (as % of Files Length) that will be filled with zeros to keep the window count constant; larger gaps cause the file to be skipped.

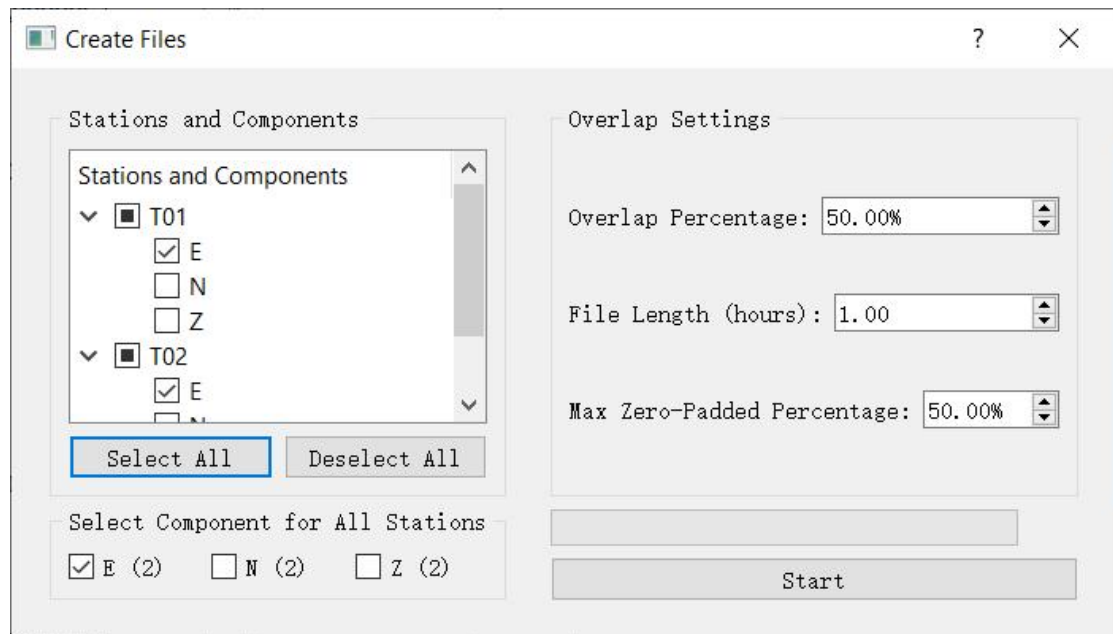


Fig.30 Overlapped files creation settings

3. Click **Start** button to create. Overlapped files will be created between the existed files.

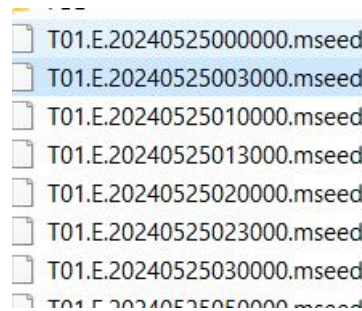


Fig.31 Overlapped files

3.2 PSD Parameters Test

1. Click **PSD - PSD Parameter Test** to open the dialog.

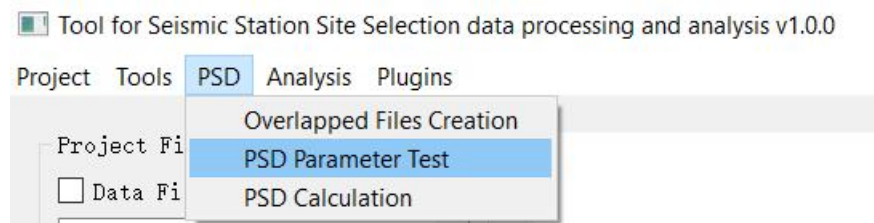


Fig.32 PSD parameter test menu

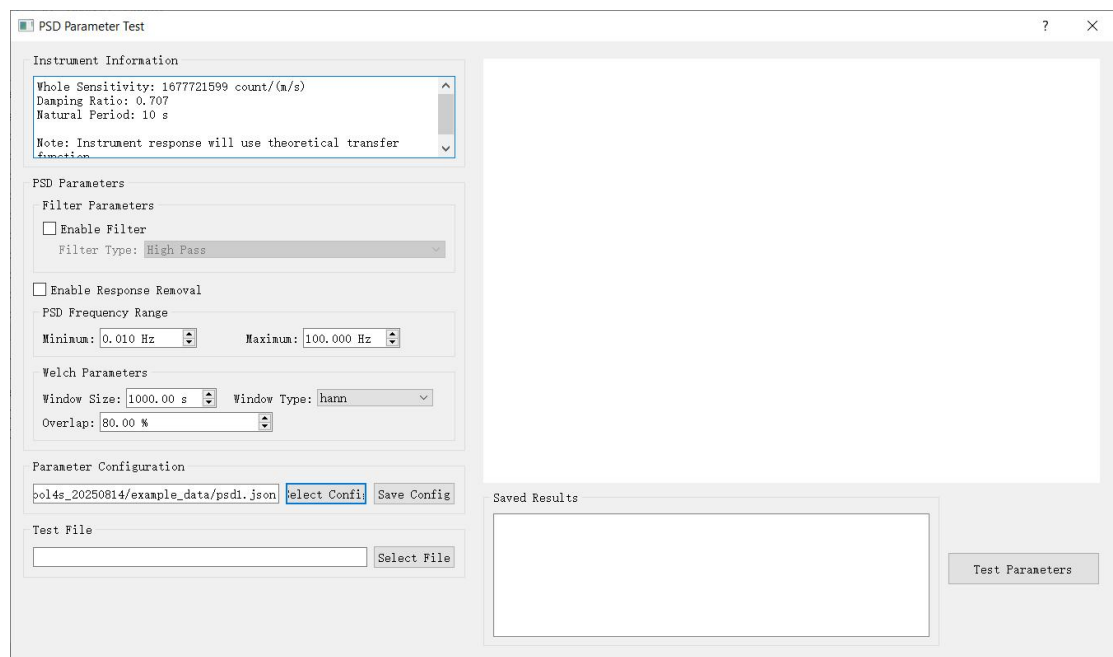


Fig.33 PSD parameter test

Instrument Information

This area displays the instrument constants that were entered in **Project Settings** (natural period, damping, whole sensitivity). They are shown here for reference and are used automatically when **Enable Response Removal** is activated.

2. Set the parameters below.

Filter Parameters

- Enable Filter: Tick to apply a causal filter before PSD computation.
- Filter Type: Drop-list High-Pass / Band-Pass.
- Frequency: Edit box; example 0.100 Hz (high-pass corner).

Response Removal

- Enable Response Removal : When **Enable Response Removal** is ticked, the instrument response will be removed based on the instrument information defined in **Project Settings**.

PSD Frequency Range

Controls the PSD frequency range.

Welch Parameters

- Window Size: Length of each Welch segment.
- Window Type: Set a suitable window
- Overlap Percentage: Overlap between successive windows.

Parameter Configuration

- Select Config: Select a psd configuration file.
- Save Config: Save the current parameters into a json file.

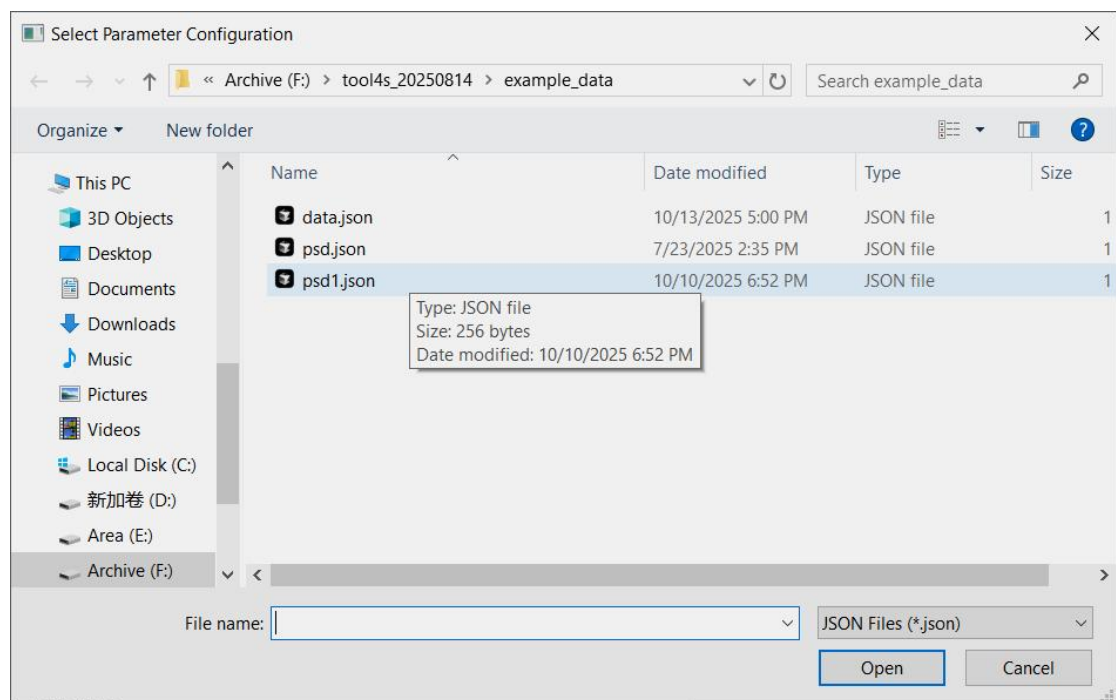


Fig.34 Select a PSD configuration file

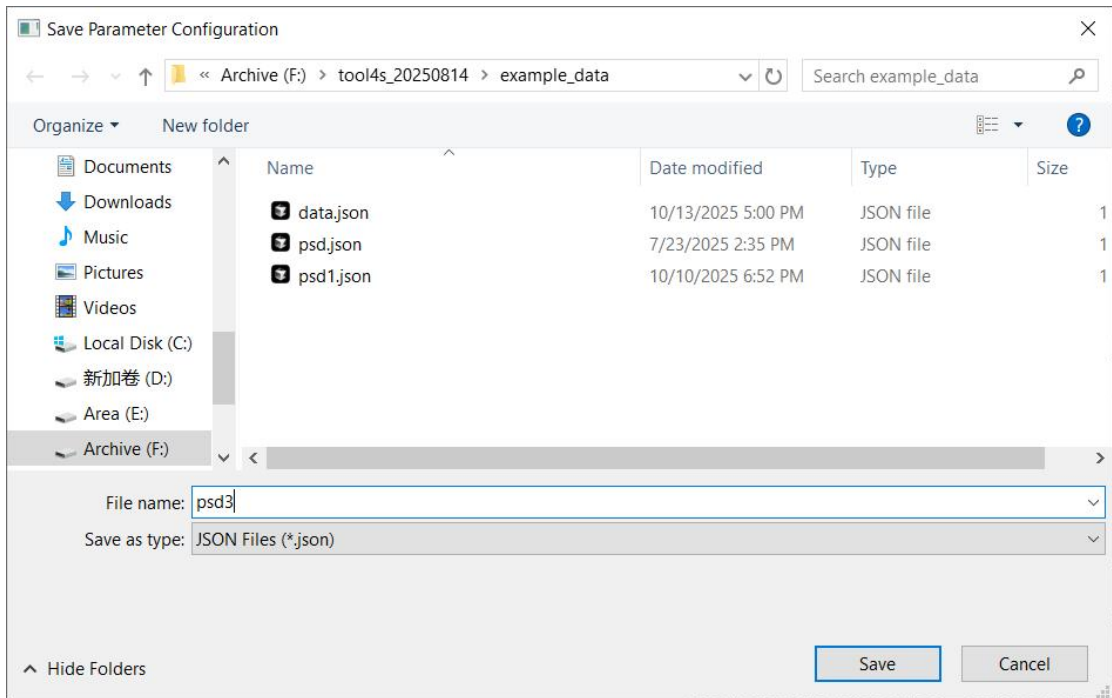


Fig.35 Select a PSD configuration file

3. Click **Select File** button to browse and select a file created by Tool4s in the output folder.

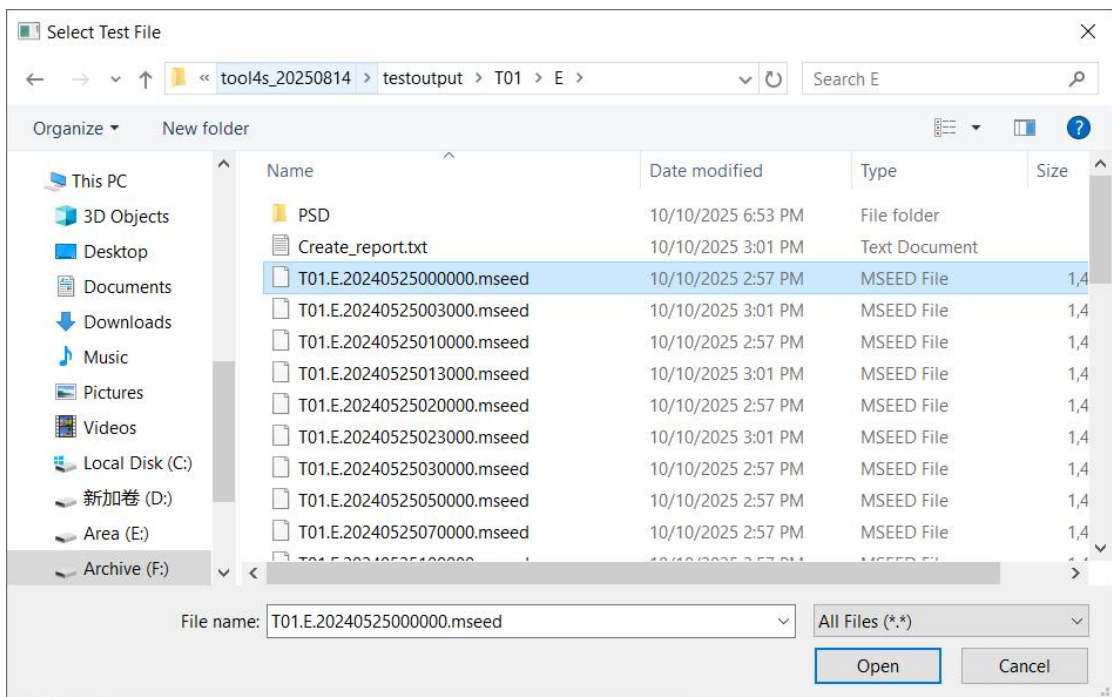


Fig.36 Save a PSD configuration file

4. Click **Test Parameters** to computes PSD instantly with the shown values.

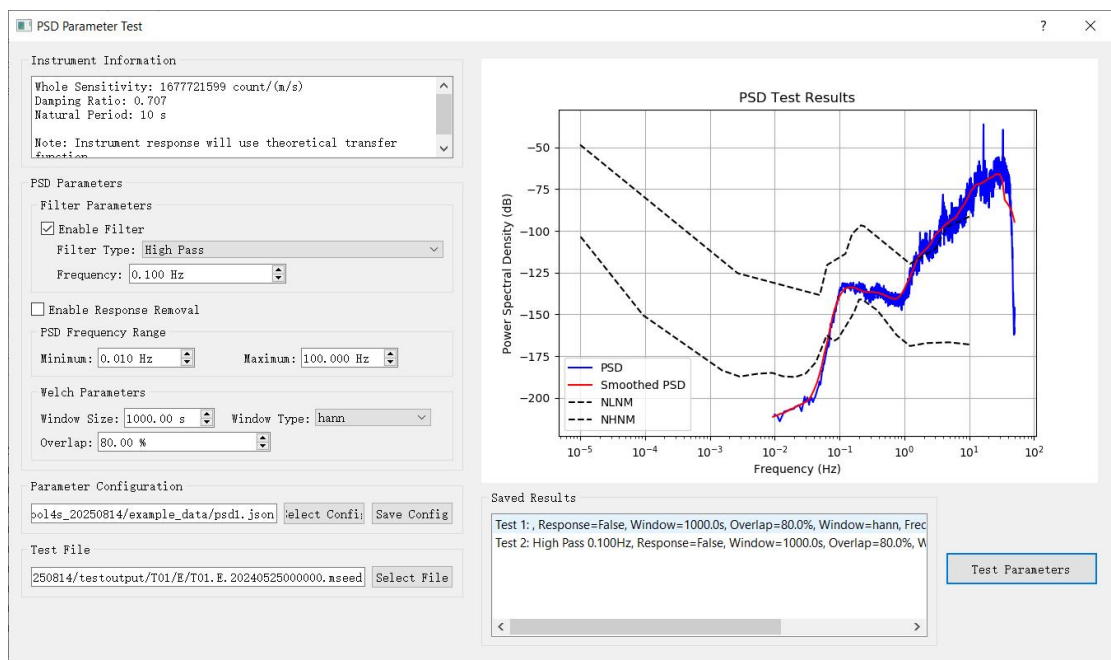


Fig.37 PSD test

3.3 PSD Calculation

1. Click **PSD - PSD Calculation** to open the dialog.

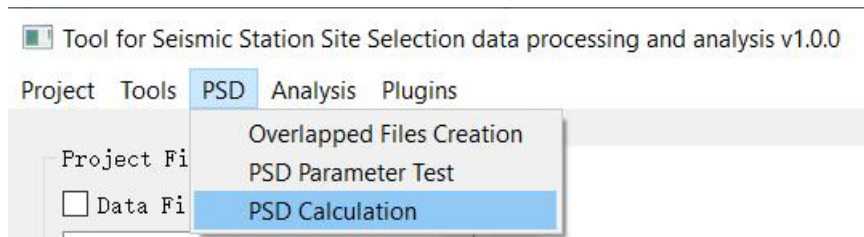


Fig.38 PSD calculation menu

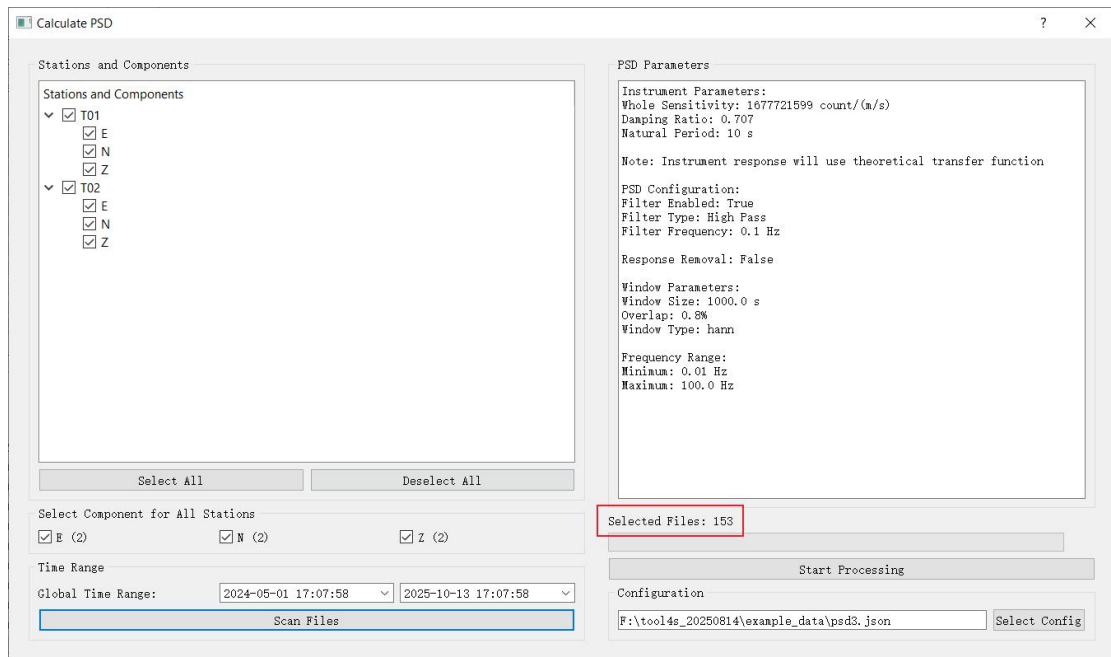


Fig.39 PSD calculation settings

2. Set the parameters below.

PSD Parameters

This area will show the instrument information.

Station and Components

Select the station and component you want to calculate PSD.

Select Component for All Stations

Select the component for all stations you want to calculate PSD.

Time Range

- Global Time Range: Use the Start Time and End Time controls to set the exact period for which PSD will be calculated.

Configuration

- Select Config: Select a PSD configuration file.
3. Click **Scan Files** button after the stations, components and time range have been set up. The **Selected Files** counter will instantly show how many files match the current conditions and will be used for the PSD calculation.
 4. Click **Start Processing** button to calculate PSD.

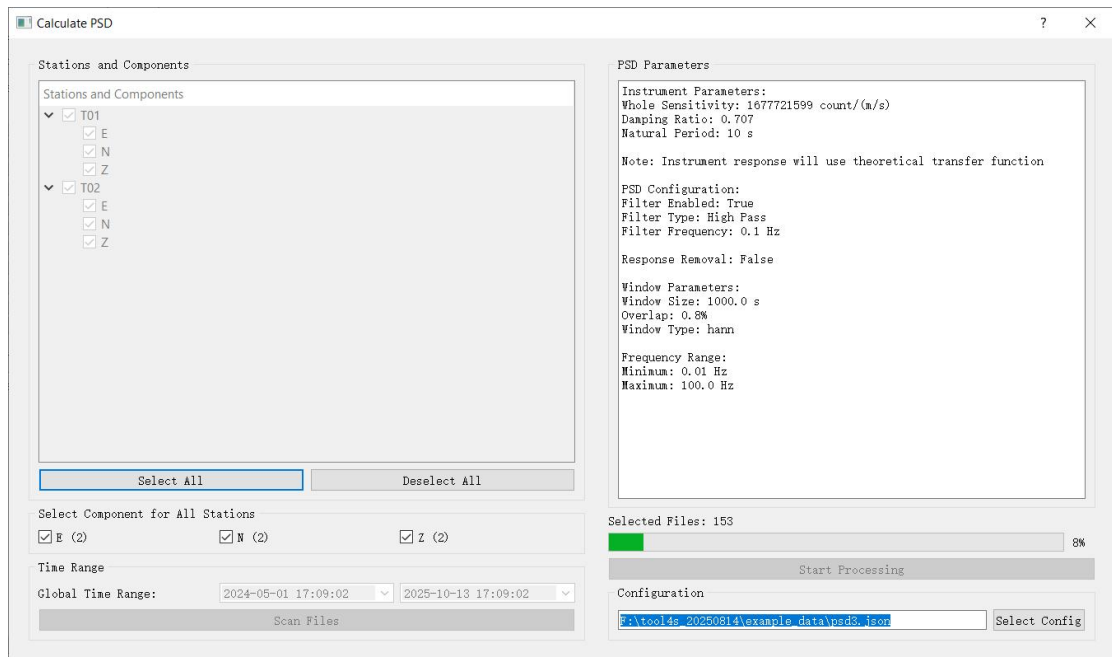


Fig.40 PSD calculation

The PSD files will be written to the PSD folder in the **Output Folder** specified in the **Project Parameters** dialog. Double-click one of the PSD file will plot the PSD line.

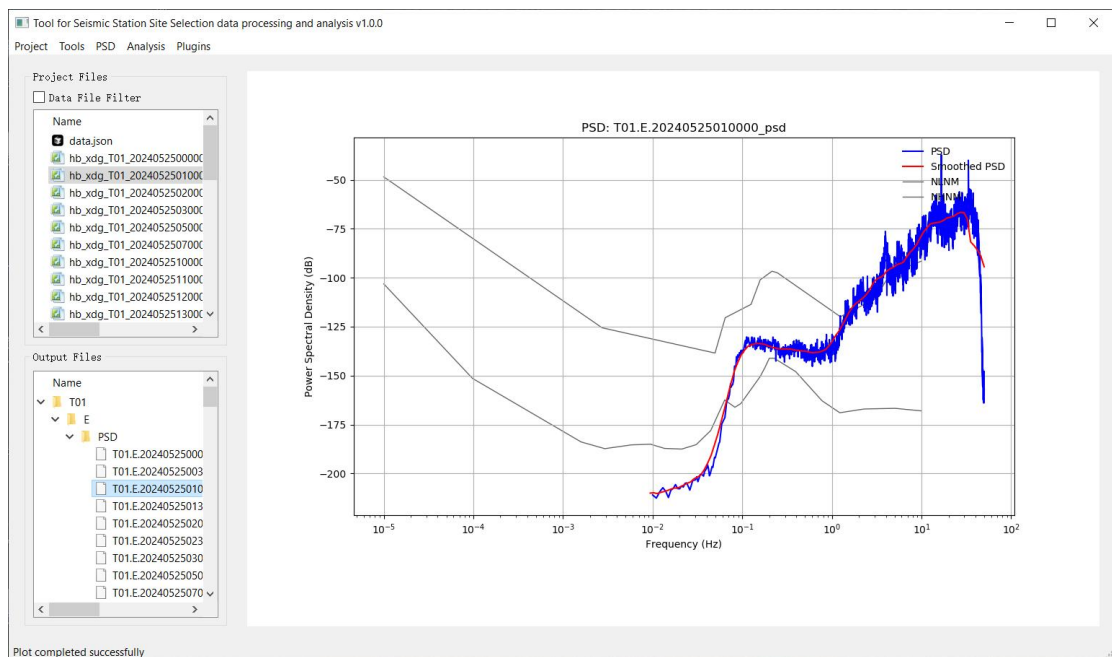


Fig.41 PSD calculation

4 Analysis

1. Click **Analysis - PSD Analysis** to open this dialog.

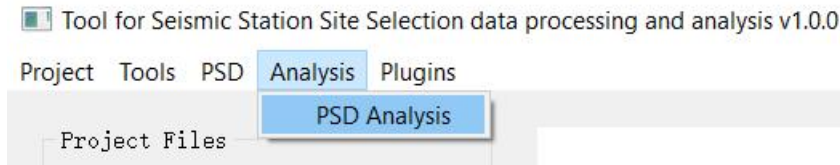


Fig.42 PSD analysis menu

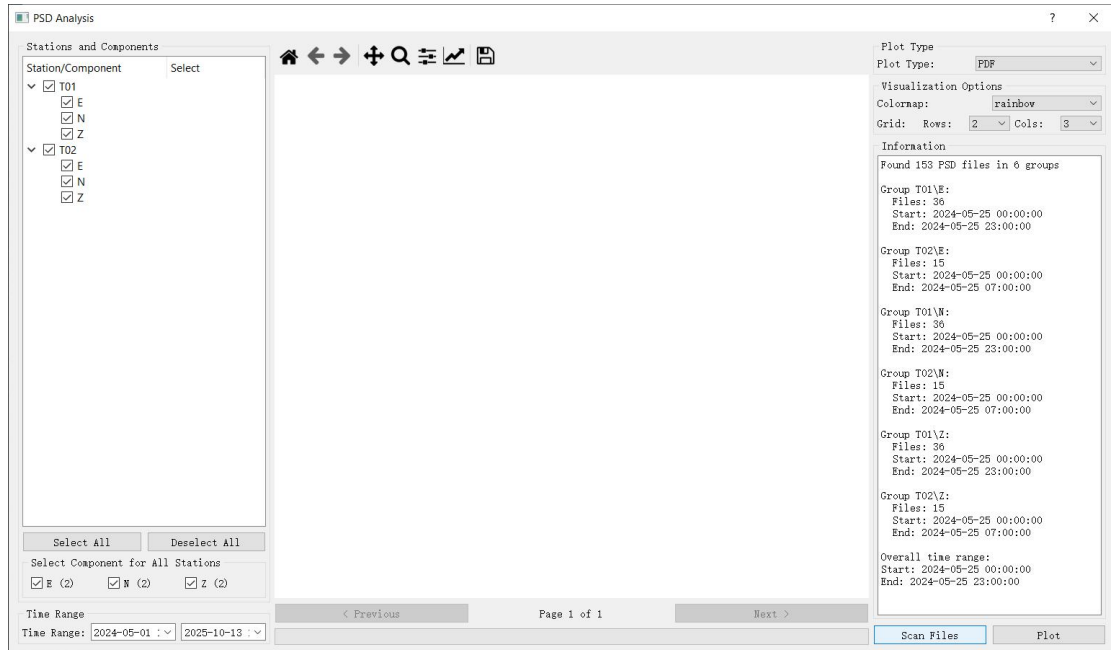


Fig.43 PSD analysis settings

2. Set the parameters below.

Station and Components

Select the station and component you want to analyze PSD.

Select Component for All Stations

Select the component for all stations you want to analyze PSD.

Time Range

- Global Time Range: Use the Start Time and End Time controls to set the exact period for which PSD will be analyzed.

Plot Type

Currently, Tool4S supports PDF, PSD, PSD time-frequency analysis.

Visualization Options

- Colormap: Colormaps from matplotlib.
- Grid: The subplots will be create by the **Rows** and **Cols**

3. Click **Scan Files** button, the **Information** area will Show the information of PSD files satisfy

current condition.

4. Click **Plot** button to plot the results.

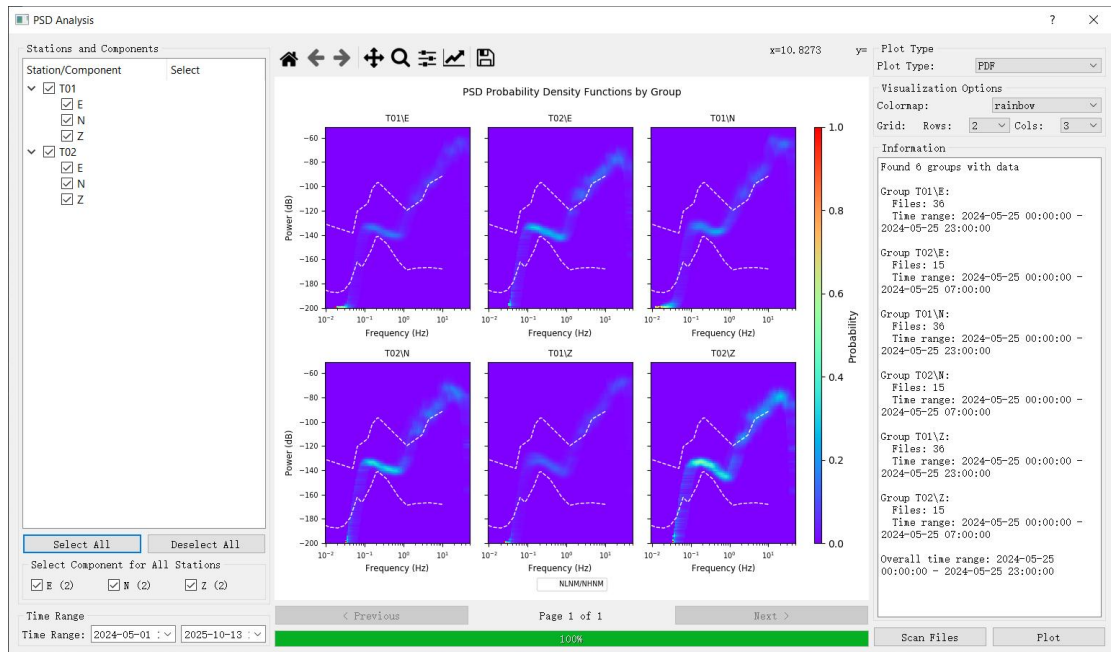


Fig.44 PDF plotting

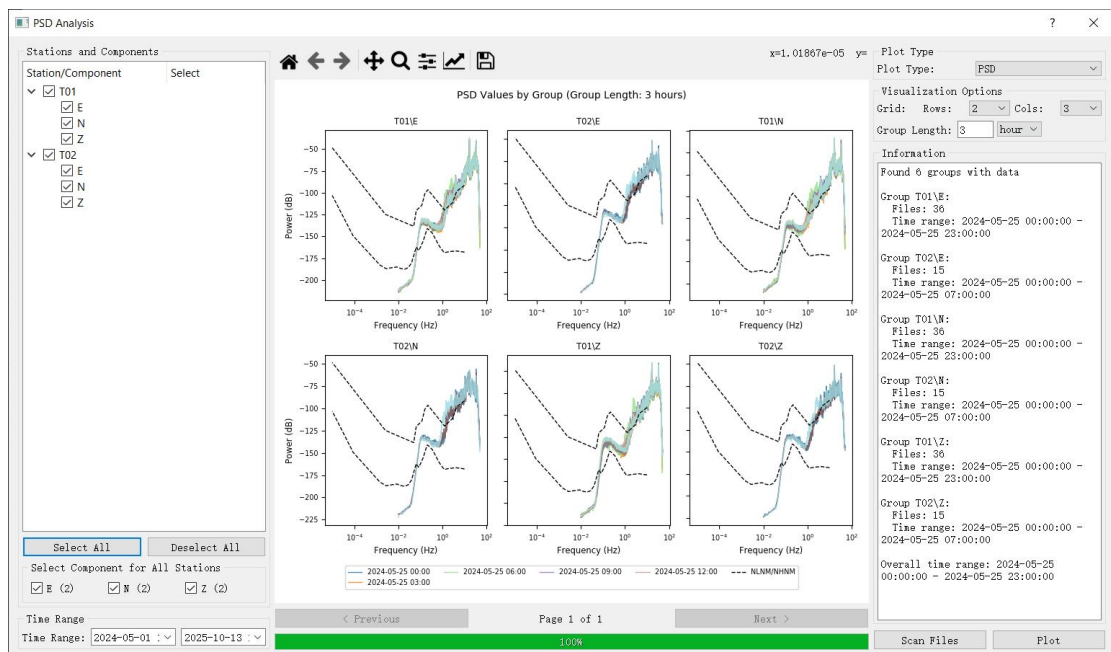


Fig.45 PSD plotting

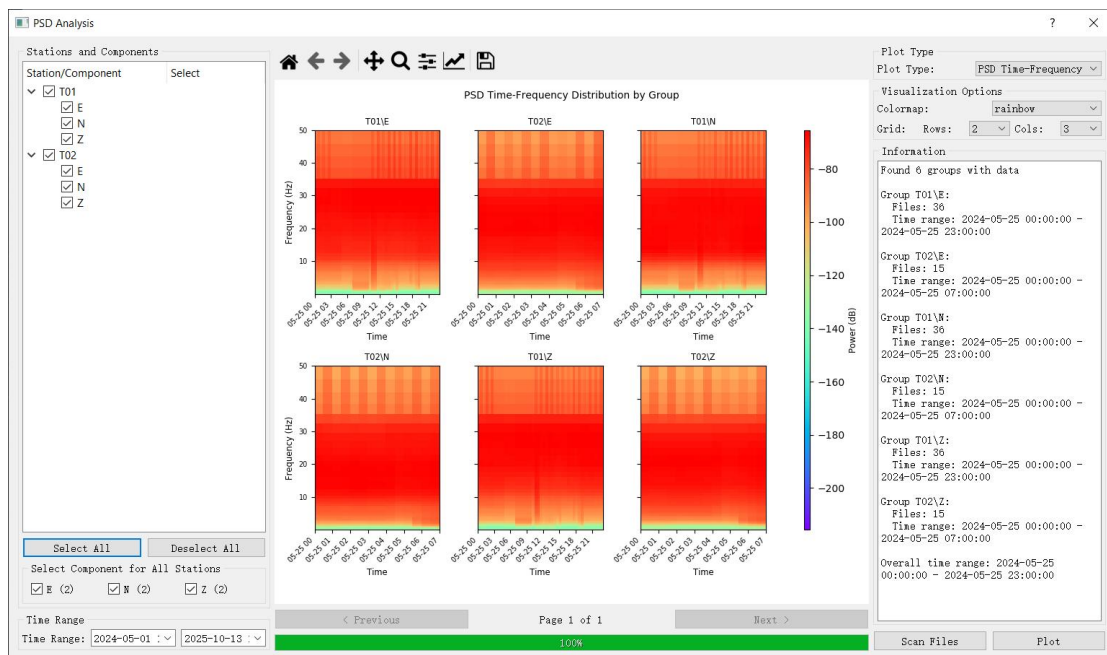


Fig.46 PSD time-frequency plotting

5 Other features

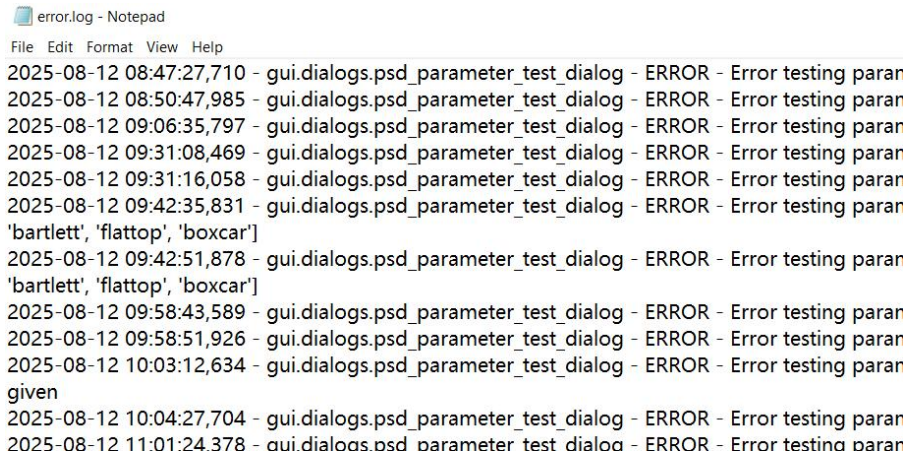
5.1 Log System

All error messages and run-time information are written to log files stored in the logs folder (located inside your Tool4S working directory).

← → ▾ ▴ 📁 > This PC > Archive (F:) > tool4s_20250814 > logs

Name	Date modified	Type	Size
error.log	10/10/2025 5:00 PM	Text Document	28 KB
tool4s.log	10/13/2025 5:12 PM	Text Document	2,090 KB
tool4s.log.1	10/9/2025 12:04 PM	1 File	10,240 KB
tool4s.log.2	10/8/2025 3:42 PM	2 File	10,240 KB
tool4s.log.3	9/25/2025 9:05 PM	3 File	10,240 KB
tool4s.log.4	9/25/2025 11:23 AM	4 File	10,240 KB
tool4s.log.5	9/25/2025 8:55 AM	5 File	10,240 KB

Fig.47 Logs

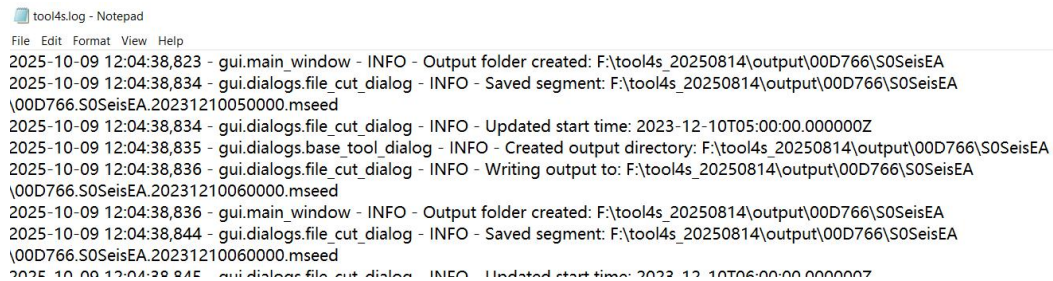


```

error.log - Notepad
File Edit Format View Help
2025-08-12 08:47:27,710 - gui.dialogs.psd_parameter_test_dialog - ERROR - Error testing paran
2025-08-12 08:50:47,985 - gui.dialogs.psd_parameter_test_dialog - ERROR - Error testing paran
2025-08-12 09:06:35,797 - gui.dialogs.psd_parameter_test_dialog - ERROR - Error testing paran
2025-08-12 09:31:08,469 - gui.dialogs.psd_parameter_test_dialog - ERROR - Error testing paran
2025-08-12 09:31:16,058 - gui.dialogs.psd_parameter_test_dialog - ERROR - Error testing paran
2025-08-12 09:42:35,831 - gui.dialogs.psd_parameter_test_dialog - ERROR - Error testing paran
'bartlett', 'flattop', 'boxcar']
2025-08-12 09:42:51,878 - gui.dialogs.psd_parameter_test_dialog - ERROR - Error testing paran
'bartlett', 'flattop', 'boxcar']
2025-08-12 09:58:43,589 - gui.dialogs.psd_parameter_test_dialog - ERROR - Error testing paran
2025-08-12 09:58:51,926 - gui.dialogs.psd_parameter_test_dialog - ERROR - Error testing paran
2025-08-12 10:03:12,634 - gui.dialogs.psd_parameter_test_dialog - ERROR - Error testing paran
given
2025-08-12 10:04:27,704 - gui.dialogs.psd_parameter_test_dialog - ERROR - Error testing paran
2025-08-12 11:01:24,378 - gui.dialogs.psd_parameter_test_dialog - ERROR - Error testing paran

```

Fig.48 Error log



```

tool4s.log - Notepad
File Edit Format View Help
2025-10-09 12:04:38,823 - gui.main_window - INFO - Output folder created: F:\tool4s_20250814\output\00D766\S0SeisEA
2025-10-09 12:04:38,834 - gui.dialogs.file_cut_dialog - INFO - Saved segment: F:\tool4s_20250814\output\00D766\S0SeisEA
\00D766.S0SeisEA.20231210050000.mseed
2025-10-09 12:04:38,834 - gui.dialogs.file_cut_dialog - INFO - Updated start time: 2023-12-10T05:00:00.000000Z
2025-10-09 12:04:38,835 - gui.dialogs.base_tool_dialog - INFO - Created output directory: F:\tool4s_20250814\output\00D766\S0SeisEA
2025-10-09 12:04:38,836 - gui.dialogs.file_cut_dialog - INFO - Writing output to: F:\tool4s_20250814\output\00D766\S0SeisEA
\00D766.S0SeisEA.20231210060000.mseed
2025-10-09 12:04:38,836 - gui.main_window - INFO - Output folder created: F:\tool4s_20250814\output\00D766\S0SeisEA
2025-10-09 12:04:38,844 - gui.dialogs.file_cut_dialog - INFO - Saved segment: F:\tool4s_20250814\output\00D766\S0SeisEA
\00D766.S0SeisEA.20231210060000.mseed
2025-10-09 12:04:38,845 - gui.dialogs.file_cut_dialog - INFO - Updated start time: 2023-12-10T06:00:00.000000Z

```

Fig.49 Run-time log

5.2 Plugins

A plugin-based design, using Python's Abstract Base Class (ABC) and the `@abstractmethod` decorator, allows adding support for new data formats. With the help of a plugin manager, data readers for different formats are loaded when the application startups. With this plugin function, it's possible to add new data readers for specific formats without modifying any code of the application. A new data format can be supported simply by adding a new data reader python file in the plugin directory without packing again. Theoretically, Tool4S can support any data format whether they are common formats or new ones from new companies.

1. Click **Plugins - Data Readers** to view current supported readers.

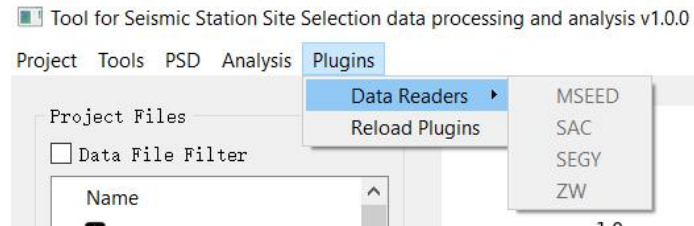


Fig.50 Data readers menu

2. Click **Reload Plugins** to reload the readers if new reader is put into the plugins folder.

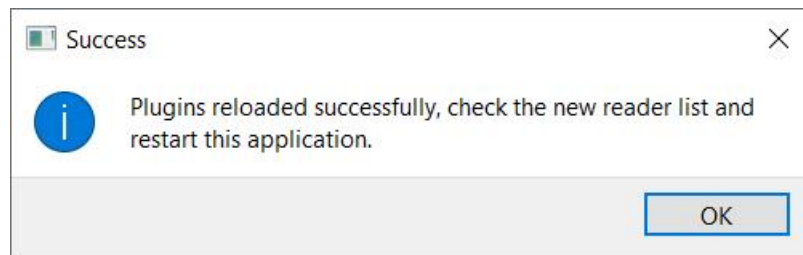


Fig.51 Reload readers